

Chapter 10.4 Conic Sections

Sketch the parabola, and label the focus, vertex, and directrix:

5. (a) $(y - 1)^2 = -12(x + 4)$

The equation represents a translated parabola whose vertex is at $(-4, 1)$.

Since $p = 3$, and the parabola opens to the left, coordinates of focus $(-7, 1)$.

The directrix is perpendicular to the axis of symmetry at a distance of $p = 3$ units from the vertex on the opposite side from the focus, so its equation is $x = -1$.

For the graph of the parabola, please see page A78 (ANSWER PART).

6. (a) $y^2 - 6y - 2x + 1 = 0$

The equation can be written in the form as

$$y^2 - 6y + 9 = 9 - 1 + 2x \Rightarrow (y - 3)^2 = 2(x + 4)$$

which is the equation of a translated parabola whose vertex is at $(-4, 3)$.

Since $p = \frac{1}{2}$, and the parabola opens to the right, coordinates of focus $(-\frac{7}{2}, 3)$.

The directrix is perpendicular to the axis of symmetry at a distance of $p = \frac{1}{2}$ unit from the vertex on the opposite side from the focus, so its equation is $x = -\frac{9}{2}$.

For graph of the parabola is left as exercise.

Sketch the ellipse, and label the foci, vertices, and ends of the minor axis:

9. (a) $(x + 3)^2 + 4(y - 5)^2 = 16$

The equation can be written in the form as

$$\frac{(x+3)^2}{16} + \frac{(y-5)^2}{4} = 1$$

which is the equation of a translated ellipse whose center is at $(-3, 5)$ and its major axis is parallel to the x -axis.

Here, $a = 4$, $b = 2$ and so $c = \sqrt{16 - 4} = 2\sqrt{3}$. So, the foci are at $(-3 - 2\sqrt{3}, 5)$, $(-3 + 2\sqrt{3}, 5)$, vertices are at $(-7, 5)$, $(1, 5)$ and endpoints of the minor axis are $(-3, 7)$ and $(-3, 3)$.

For the graph of the ellipse, please see page A79 (ANSWER PART).

(b) $\frac{1}{4}x^2 + \frac{1}{9}(y+2)^2 - 1 = 0$

The equation can be written in the form as

$$\frac{x^2}{4} + \frac{(y+2)^2}{9} = 1$$

which is the equation of a translated ellipse whose center is at $(0, -2)$ and its major axis is parallel to the y -axis.

Here, $a = 3$, $b = 2$ and so $c = \sqrt{9 - 4} = \sqrt{5}$. So, the foci are at $(0, -2 + \sqrt{5})$, $(0, -2 - \sqrt{5})$, vertices are at $(0, 1)$, $(0, -5)$ and endpoints of the minor axis are $(-2, -2)$ and $(2, -2)$.

For the graph of the ellipse, please see page A79 (ANSWER PART).

Sketch the hyperbola, and label the vertices, foci, and asymptotes.

13. (a) $\frac{(y+4)^2}{3} - \frac{(x-2)^2}{5} = 1$

This is the equation of a translated hyperbola whose center is at $(2, -4)$ and its major axis is parallel to the y -axis.

Here, $a = \sqrt{3}$, $b = \sqrt{5}$ and so $c = \sqrt{3 + 5} = 2\sqrt{2}$.

So, the foci are at $(2, -4 + 2\sqrt{2})$, $(2, -4 - 2\sqrt{2})$ and the vertices are at $(2, -4 + \sqrt{3})$, $(2, -4 - \sqrt{3})$.

The asymptotes of the hyperbola are given by

$$\begin{aligned} \frac{(y+4)^2}{3} - \frac{(x-2)^2}{5} &= 0 \Rightarrow 5(y+4)^2 = 3(x-2)^2 \\ \Rightarrow y+4 &= \pm \sqrt{\frac{3}{5}}(x-2) \end{aligned}$$

For the graph of the hyperbola, please see page A79 (ANSWER PART).

(b) $16(x+1)^2 - 8(y-3)^2 = 16$

The equation can be written in the form as

$$\frac{(x+1)^2}{1} - \frac{(y-3)^2}{2} = 1$$

which is the equation of a translated hyperbola whose center is at $(-1, 3)$ and its major axis is parallel to the x -axis.

Here, $a = 1$, $b = \sqrt{2}$ and so $c = \sqrt{1 + 2} = \sqrt{3}$.

So, the foci are at $(-2, 3)$, $(0, 3)$ and the vertices are at $(-1 - \sqrt{3}, 3)$, $(-1 + \sqrt{3}, 3)$.

The asymptotes of the hyperbola are given by

$$\frac{(x+1)^2}{1} - \frac{(y-3)^2}{2} = 0 \Rightarrow (y-3)^2 = 2(x+1)^2$$

$$\Rightarrow y-3 = \pm\sqrt{2}(x+1)$$

For the graph of the hyperbola, please see page A79 (ANSWER PART).

Chapter 10.6 Conic Sections in Polar Coordinates

All of the important geometric information about an ellipse can be obtained from the values of a , b , and c . One way to find these values from the polar equation of an ellipse is based on finding the distances from the focus to the vertices.

Let r_0 be the distance from the focus to the closest vertex and r_1 the distance to the farthest vertex. Thus, we have

$$r_0 = a - c \text{ and } r_1 = a + c \Rightarrow r_0 r_1 = a^2 - c^2 = b^2 \Rightarrow b = \sqrt{r_0 r_1}$$

$$\text{and } a = \frac{1}{2}(r_1 + r_0), \quad c = \frac{1}{2}(r_1 - r_0).$$

Please see Figure 10.6.5

Please See Example 2

Find the distances from the pole to the vertices, and then apply Formulas to find the equation of the ellipse in rectangular coordinates.

7. (a) $r = \frac{6}{2+\sin\theta}$

Rewrite the polar equation in the form as

$$r = \frac{6}{2+\sin\theta} \Rightarrow r = \frac{3}{1+\frac{1}{2}\sin\theta}$$

which is an equation of an ellipse.

The distance r_0 from the focus to the closest vertex can be obtained by setting $\theta = 0$ in this equation, and the distance r_1 to the farthest vertex can be obtained by setting $\theta = \pi$. This yields

$$r_0 = \frac{6}{2+\sin\frac{\pi}{2}} = 2, \quad r_1 = \frac{6}{2+\sin\frac{3\pi}{2}} = \frac{6}{2-\sin\frac{\pi}{2}} = 6$$

Therefore, we get $b = \sqrt{(2)(6)} = \sqrt{12} = 2\sqrt{3}$ and $a = \frac{1}{2}(6+2) = 4$, $c = \frac{1}{2}(6-2) = 2$.

In this case, the directrix is parallel to the polar axis, i.e. it is above the pole.

Hence, the equation of the ellipse in rectangular coordinates is

$$\frac{x^2}{12} + \frac{(y+2)^2}{16} = 1$$

For the ellipse and the hyperbola, we will take the directrix to be $x = \frac{a^2}{c}$.

Please see Figure 10.6.1

We will assume that the focus is at the pole and the directrix is either parallel or perpendicular to the polar axis. If the directrix is parallel to the polar axis, then it can be above or below the pole; and if the directrix is perpendicular to the polar axis, then it can be to the left or right of the pole.

Please see Figure 10.6.3

Theorem 10.6.2

Shapes of elliptical orbits are often specified by giving the eccentricity e and the semi-major axis a , so it is useful to express the polar equations of an ellipse in terms of these constants.

The relationship $c = ae$ implies that the distance d between the focus and the directrix is

$$d = \frac{a}{e} - c = \frac{a}{e} - ae = \frac{a(1 - e^2)}{e}$$

Please see Figure 10.6.11

All of the important geometric information about a hyperbola can be obtained from the values of a , b , and c . As with the ellipse, one way to find these values from the polar equation of a hyperbola is based on finding the distances from the focus to the vertices.

Let r_0 be the distance from the focus to the closest vertex and r_1 the distance to the farthest vertex. Thus, we have

$$r_0 = c - a \quad \text{and} \quad r_1 = c + a \quad \Rightarrow \quad r_0 r_1 = c^2 - a^2 = b^2 \Rightarrow b = \sqrt{r_0 r_1}$$

$$\text{and} \quad a = \frac{1}{2}(r_1 - r_0), \quad c = \frac{1}{2}(r_1 + r_0).$$

Please see Figure 10.6.7

Please See Example 3